

MAJOR PROJECT - PHASE II



MedGuard AI : Intelligent System For Medicine Quality Check using Agentic AI

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1. Introduction & Problem Statement

Introduction :

- Growing rise of counterfeit & expired medicines impacts patient safety and trust.
- Existing verification methods are manual, slow, and error-prone.
- MedGuard AI provides real-time, AI-driven verification using image, text, and barcode inputs.

Problem Statement : Need for an automated, reliable, and accessible system to authenticate medicines using multi-modal inputs and AI agents in real-world consumer environments.



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2. Modules Proposed in Project Phase-I

User Input Layer

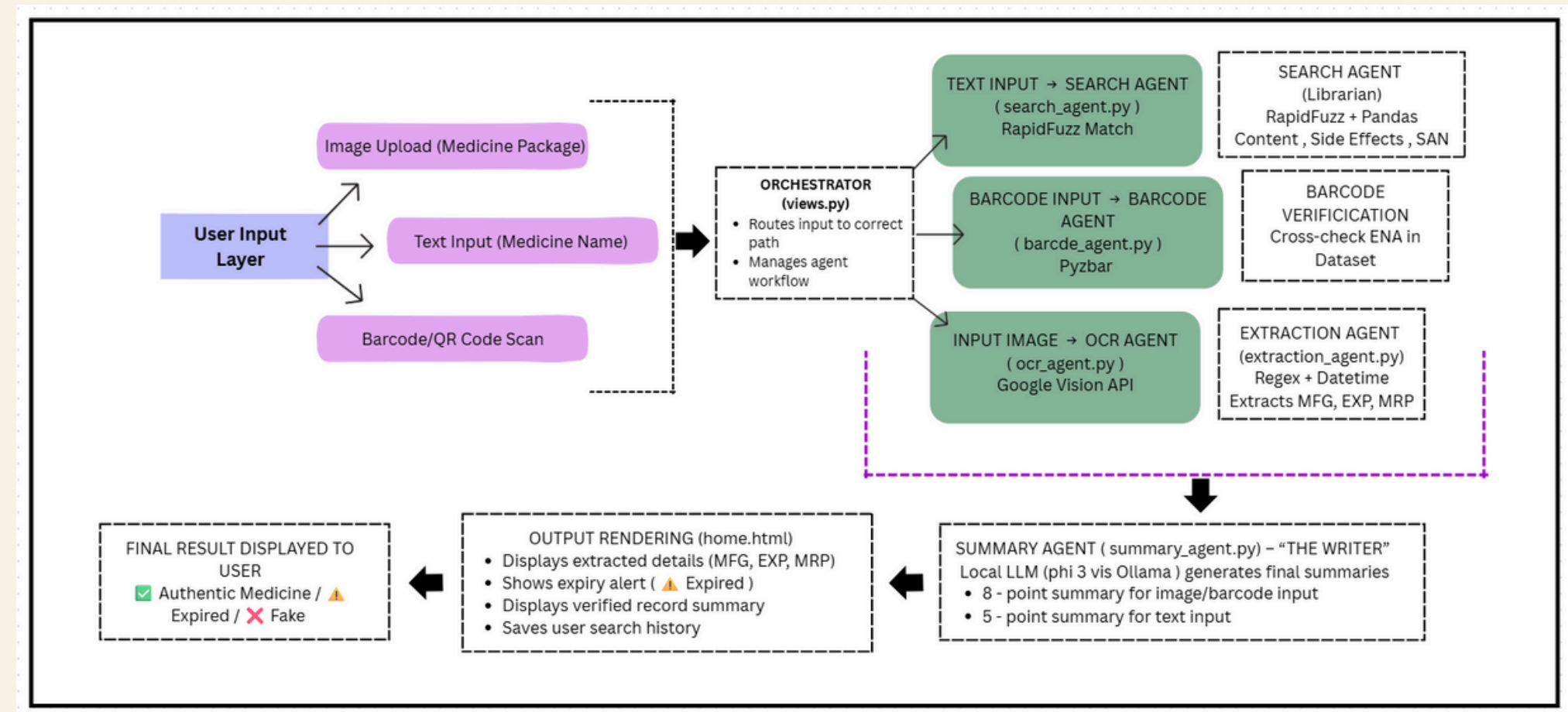
- Text Input
- Image Upload
- Barcode Scan

AI Agent Modules

- OCR Agent (Google Vision)
- Barcode Agent (pyzbar)
- Extraction Agent
- (Regex for MFG/EXP/MRP)
- Search Agent (RapidFuzz + Dataset)
- Summary Agent (Local LLM via Ollama)

Verification & Output Layer

- Expiry & Authenticity Check
- User History & Results Storage



3. Status of Modules Completed in Phase-I

- Multi-modal Inputs (Text, Image, Barcode)
- AI Agent Workflow implemented
- Expiry & Authenticity detection enabled
- Local Dataset of 500+ medicines integrated
- Login + User History functionality added
- Front-end web interface deployed (Django-based)
- Local LLM summary generation (phi-3 mini via Ollama)



4. Modules Proposed for Project Phase-II

- ◆ **UI/UX Enhancement**
 - React.js + Tailwind + Framer Motion for modern professional UI
- ◆ **Mobile Application Integration**
 - Flutter app for Android
 - Camera-based image capture
 - On-device Barcode Scan (Google ML Kit)
- ◆ **Backend Transformation**
 - Convert Django to Headless API
 - Django REST Framework + JWT Auth
 - Unified PostgreSQL Database
- ◆ **Cross-Platform Sync**
 - Shared history & data across Web + Mobile login
- ◆ **Cloud Deployment**
 - Backend: AWS / DigitalOcean
 - Web: Vercel / Netlify
 - Mobile: Play Store Distribution



Modules Proposed for Project Phase-II

◆ UI/UX Enhancement

- React.js + Tailwind + Framer Motion for modern professional UI

◆ Mobile Application Integration

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Phase-1 Completed

- Core Agentic AI pipeline (OCR + Barcode + LLM)
- Web app & user history module

Phase-2 Scheduled

1. API Layer (DRF + JWT)
2. Flutter Mobile App + Camera + Barcode
3. React Front-End (1mg-style UI)
4. Cloud Deployment + Testing
5. Play Store Release
6. Performance & UX optimization



7. Status of Publication

Research paper has been successfully submitted for review. for Q1.